

Nanda Kumar Mangati

Senior Full-Stack Engineer

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SUMMARY

Full-stack engineer with **11+ years** owning the full software lifecycle, from requirements and architecture through deployment and production support, for multi-tenant SaaS and internal platforms. I spent a decade building a **customer-support SaaS suite** (enterprise Helpdesk, Workflows automation, BI) serving 5,000+ businesses: I built and scaled its integration layer and led the team behind its internal iPaaS platform. I design extensible REST APIs and backend services, lead architectural reviews, and mentor engineers, with a focus on backend scalability, database and browser-rendering performance, high availability, and multi-tenant fault tolerance. AI coding agents (Claude Code, Codex) are part of my daily workflow.

SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL, HTML/CSS
- **Backend & APIs:** Django, Django Ninja, Node.js, REST API design, microservices, distributed task queues (Celery), background jobs, auth & SSO (OIDC, JWT, Auth.js/next-auth v5)
- **Data:** RDBMS (PostgreSQL), NoSQL (MongoDB, Redis), ElasticSearch, Drizzle ORM, Turso/libSQL; query optimization
- **Frontend:** React 19 (Server Components, Server Actions), Next.js 16, Ember.js, state management (TanStack Query/Table/Virtual), Radix UI, Tailwind CSS, React Hook Form, Zod, Vite, Webpack
- **Architecture:** multi-tenant SaaS, high availability, fault tolerance, caching, security (fail-closed authorization, tenant isolation, SSO), full-lifecycle ownership
- **Cloud & DevOps:** AWS (S3, Lambda, CloudWatch), Docker, CI/CD (GitHub Actions), Nginx
- **Testing:** pytest, Jest, MSW, Playwright (e2e), Storybook
- **AI:** AI coding agents (Claude Code, Codex), LLM integration, RAG, MCP (writing at mangatinanda.me/blog)

EXPERIENCE

Isha Foundation (IT) · Full Stack Developer

Jul 2025 – Present

- Full-stack contributor on **IVM**, a ground-up production volunteer-management platform running ~20 programs (Django 6 / Django Ninja REST API, Next.js 16 volunteer PWA, admin dashboard).
- **Own the volunteer-facing PWA** as its primary author: I set its architecture, designed the SSO auth model that both frontends now share, and hold the release bar with automated quality gates on every change, capturing key decisions as ADRs.
- On the backend, **designed and shipped the volunteer duty-rotation engine end to end**, from data model through API surface to its nightly scheduled runs, and tuned scheduling queries on the hot path.

HappyFox Technologies LLP · Full Stack Developer

2021 – Jun 2025

Multi-tenant customer-support SaaS: Enterprise Helpdesk, Workflows automation, and BI products.

- Led **4 developers and a QA engineer** to build an internal **iPaaS platform** for building and shipping integrations across HappyFox products; it **cut integration development time by 50%** and had **3 products onboarded** when I moved on. I owned its frontend and worked hands-on in the Python/Django backend.
- **Built and scaled the core integration layer** connecting Helpdesk, Workflows, and BI to each other and to a catalog of **40+ third-party integrations** used by tens of thousands of support agents: I designed the REST APIs, database schemas, and backend services behind it.

- **Led design reviews** for integration features and **mentored junior developers** on architecture, coding practices, and design.
- Built resilient two-way integrations (Jira, Asana, Facebook, X) with retries and backoff, and real-time webhook ingestion for RingCentral, debugging production issues through AWS CloudWatch.
- Ran long-running integration jobs as background work, on Helpdesk's priority-based task infrastructure and on **Celery pipelines** in Workflows, keeping heavy processing out of the request path.
- On the **BI product**, built dashboards, integrations, and query optimizations over high-volume ticket data (millions of tickets).

HappyFox Technologies LLP · Front-End Developer

May 2015 – 2021

- Key contributor to the full UI revamp of the enterprise helpdesk into a **performant Ember.js single-page app**, shipped via weekly-to-daily CI/CD. I tuned component lifecycle and client-side state management for large, data-dense views.
- Designed and built the reusable **JavaScript SDK** that became the foundation of third-party integrations for the core helpdesk platform.
- Built integrations with third-party platforms including **Salesforce, Shopify, Microsoft Dynamics 365, and RingCentral**, spanning project management, CRM, e-commerce, social media, and voice.

Cognizant Technology Solutions · Programmer Analyst

Dec 2013 – May 2015

- Built reusable front-end templates and fixed UI defects on an Ember.js online-banking application, after formal web-development training.

SELECTED PROJECTS & WRITING

Outlay · collaborative multi-tenant household expense-tracking PWA · outlay.mangatinanda.me

Server-first Next.js 16 (React 19 Server Components + Server Actions) · Drizzle ORM + Turso/libSQL · Auth.js v5 · TypeScript (strict) · built solo with AI coding agents · live with daily users

- **Multi-tenant isolation by construction:** every write re-resolves the active tenant on the server and filters on `household_id`, so a browser cannot forge cross-tenant access. A dedicated end-to-end test switches tenants and fails CI the instant one tenant can read another's data.
- **Production rigor as a solo maintainer:** 227 tests across 48 files (27 integration tests run the real Drizzle migrations on in-memory libSQL, no mocks) behind a blocking CI gate: lint, typecheck, full suite, production build.

mangatinanda.me · personal portfolio and technical blog

Next.js 16 · writing on MCP, RAG, and AI-assisted engineering

EDUCATION

B.Tech, Computer Science & Engineering, Sri Venkateswara University College of Engineering, Tirupati, 2009 – 2013

Languages: Telugu, English, Tamil, Hindi